

Yossi Abutbul

Computer Science Student | RF Integrator & Test Automation
052-547-6603 · abyossi22@gmail.com · [GitHub](#) · [LinkedIn](#) · [Portfolio](#)

SUMMARY

CS student and RF Integrator with 6+ years of hands-on experience. Builds RF test-automation platforms and antenna visualisation tools across the full stack (Python/FastAPI, React/Next.js, TypeScript). Experienced in NB-IoT, LoRa, LTE, and Bluetooth bring-up and signal analysis. Delivered tooling that cut multi-day manual tests to minutes.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · C

Frontend: React · Vite · Chart.js · Plotly · Responsive UI · Next.js · Tailwind

Backend: FastAPI · REST APIs · Firebase · Firestore · Python Automation

RF / Embedded: Spectrum Analyzer · Network Analyzer · NB-IoT · CAT-M · LoRa · LTE · Bluetooth

Tools: Git · Claude · Codex · n8n

WORK EXPERIENCE

RF & Electronics Integrator - Arad Technologies

May 2020 – Present

- Designed a React + FastAPI platform automating RF tests over Bluetooth with Power & Spectrum Analyzers - **cut lab reporting time by 50%+.**
- Built an automated load-pull test system (DC Power Analyzer, Network Analyzer, Power Sensor, motor-driven trombone tuner, Arduino-controlled RF switch on a custom 3D-printed rig) – **from 3-day manual test to ~8 minutes automated.**
- Built an interactive web tool for **visualising antenna radiation patterns** (2D polar & 3D surface) and RF performance metrics.
- Led HW/SW **bring-up** of NB-IoT, CAT-M, and LoRa modules with FEM integration and Python-based validation.

MILITARY SERVICE

Operational Project Leader & RF Technician - IDF Intelligence Corps – Unit 81

Apr 2017 – Dec 2019

- Led multi-disciplinary RF/tech projects under tight operational deadlines.
- Built Python tools to log and analyse Spectrum Analyzer data in the field.
- Hands-on troubleshooting and integration of advanced RF systems.

PROJECTS

OPlanner | React · TypeScript · Firebase

Full-stack semester and task planner with reusable components, Firebase-backed state, and progress dashboards.

RF Report Generator | React · TypeScript · Python · FastAPI · Plotly

Captures live instrument measurements and generates structured reports with 2D polar & 3D radiation visualisations. Cut manual reporting time by 50%+.

Haparlamentor | React · TypeScript · Python · OpenAI Whisper

Hebrew phrase-to-episode search for the Israeli sitcom “**Haparlament**”. Fuzzy search across transcripts, deep links to the exact Mako episode. Built a custom agent that scrapes episodes and transcribes them with OpenAI Whisper to make the whole thing possible.

Pipeline CPU Simulator | React · TypeScript · Vite

Educational CPU pipeline visualiser - step through instructions and observe hazard detection and data forwarding.

EDUCATION

BSc Computer Science - The Open University of Israel · GPA 86

Oct 2022 – Present